

From	To	Electronic Value	Description
Regulated DC Supply (+)	Central Star Positive Bus (+)	5.0VDC (1.0A max)	Main positive DC feed from external supply.
Regulated DC Supply (-)	Central Star Ground Bus (-)	0V / GND	Master ground return line for the entire system.
Central Star Positive Bus (+)	Decoupling Buffer (+)	5.0VDC (470 uF 100 nF)	High-frequency and transient voltage stabilization.
Central Star Ground Bus (-)	Decoupling Buffer (-)	0V / GND	Return path for decoupling filter network.
Central Star Positive Bus (+)	Arduino UNO Pin +5V	5.0VDC (~60 mA)	Direct ATmega328P logic board supply rail.
Central Star Ground Bus (-)	Arduino UNO Pin Ground	0V / GND	Digital logic reference ground return.
Central Star Positive Bus (+)	ULN2003 #1 Pin +	5.0VDC (~180 mA max)	Driver 1 motor common and flyback diode power.
Central Star Ground Bus (-)	ULN2003 #1 Pin -	0V / GND	Driver 1 Darlington emitter ground return.
Central Star Positive Bus (+)	ULN2003 #2 Pin +	5.0VDC (~180 mA max)	Driver 2 motor common and flyback diode power.

Central Star Ground Bus (-)	ULN2003 #2 Pin -	0V / GND	Driver 2 Darlington emitter ground return.
ULN2003 #1 Jumper Pin A	ULN2003 #1 Jumper Pin B	0 Ohm Shunt	Connects +5V to Pin 9 clamp diodes and LEDs.
ULN2003 #2 Jumper Pin A	ULN2003 #2 Jumper Pin B	0 Ohm Shunt	Connects +5V to Pin 9 clamp diodes and LEDs.
Central Star Positive Bus (+)	10k Resistor Lead 1	5.0VDC	Bias source for hardware interrupt pull-up.
10k Resistor Lead 2	Arduino UNO Pin D2 (INT0)	5.0VDC / 10 kOhm	Holds interrupt pin HIGH when the button is open.
Arduino UNO Pin D2	Push Button Terminal 1	5.0V / 0V Logic	Routes active-LOW trigger on button press.
Push Button Terminal 2	Central Star Ground Bus (-)	0V / GND	Ground reference completing switch closure.
Arduino UNO Pin D2	100nF Cap Lead 1	100 nF Filter	Shunts mechanical contact bounce to ground.
100nF Cap Lead 2	Central Star Ground Bus (-)	0V / GND	Ground return for RC debounce filter network.
Arduino UNO Pin D10	ULN2003 #1 Pin IN1	5.0V Logic (~1.3 mA)	Logic drive for Motor 1 Phase A.
Arduino UNO Pin D3	ULN2003 #1 Pin IN2	5.0V Logic (~1.3 mA)	Logic drive for Motor 1 Phase B.

Arduino UNO Pin D4	ULN2003 #1 Pin IN3	5.0V Logic (~1.3 mA)	Logic drive for Motor 1 Phase C.
Arduino UNO Pin D5	ULN2003 #1 Pin IN4	5.0V Logic (~1.3 mA)	Logic drive for Motor 1 Phase D.
Arduino UNO Pin D6	ULN2003 #2 Pin IN1	5.0V Logic (~1.3 mA)	Logic drive for Motor 2 Phase A.
Arduino UNO Pin D7	ULN2003 #2 Pin IN2	5.0V Logic (~1.3 mA)	Logic drive for Motor 2 Phase B.
Arduino UNO Pin D8	ULN2003 #2 Pin IN3	5.0V Logic (~1.3 mA)	Logic drive for Motor 2 Phase C.
Arduino UNO Pin D9	ULN2003 #2 Pin IN4	5.0V Logic (~1.3 mA)	Logic drive for Motor 2 Phase D.
ATmega328P Pin PB5 (D13)	Built-in LED (LED_BUILTIN)	5.0V (~5.6 mA)	Onboard PCB trace driving interrupt status LED.
ULN2003 #1 JST Pin 1	Motor 1 Blue Wire	Switched GND (~85 mA)	Low-side sink for Motor 1 Phase A coil.
ULN2003 #1 JST Pin 2	Motor 1 Pink Wire	Switched GND (~85 mA)	Low-side sink for Motor 1 Phase B coil.
ULN2003 #1 JST Pin 3	Motor 1 Yellow Wire	Switched GND (~85 mA)	Low-side sink for Motor 1 Phase C coil.
ULN2003 #1 JST	Motor 1 Orange	Switched GND (~85	Low-side sink for

Pin 4	Wire	mA)	Motor 1 Phase D coil.
ULN2003 #1 JST Pin 5	Motor 1 Red Wire	5.0VDC (~180 mA max)	Common center-tap +5V feed for Motor 1 coils.
ULN2003 #2 JST Pin 1	Motor 2 Blue Wire	Switched GND (~85 mA)	Low-side sink for Motor 2 Phase A coil.
ULN2003 #2 JST Pin 2	Motor 2 Pink Wire	Switched GND (~85 mA)	Low-side sink for Motor 2 Phase B coil.
ULN2003 #2 JST Pin 3	Motor 2 Yellow Wire	Switched GND (~85 mA)	Low-side sink for Motor 2 Phase C coil.
ULN2003 #2 JST Pin 4	Motor 2 Orange Wire	Switched GND (~85 mA)	Low-side sink for Motor 2 Phase D coil.
ULN2003 #2 JST Pin 5	Motor 2 Red Wire	5.0VDC (~180 mA max)	Common center-tap +5V feed for Motor 2 coils.